

EE P 564: TinyML

Lab 8: American Sign Language Recognition

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ELECTRICAL & COMPUTER
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Today's Lab

- Both Software and Hardware Lab!
- Training models to recognize the alphabet
- Quantize the model for TinyML
- Deploy the model to the Arduino board



Lab 8 - Software - Road Map

1. Gesture Based ASL recognition
2. Model training
3. Model Evaluation
4. Covert the TensorFlow model to TensorFlow Lite Micro model
5. Compress the model



Lab 8 - Hardware - Road Map

1. Connect Arduino Boards to your computer
2. Implement ASL recognition using compressed ML model
3. Deploy the model on to board
4. Open Serial Monitor and test the model



What you need

- Mac, Linux or Windows laptop
- Arduino Nano BLE Sense 33
- MicroUSB cable
- Possibly a USB C to USB A converter, if you're on a USB C laptop
- Optional: Wearable Glove and Sticky tape

Source: <https://towardsdatascience.com/k-means-a-complete-introduction>



Training and Quantize the Model

Open EEP595-TinyML-Lab8-Gesture.ipynb (Gesture Based ASL) in Colab

Let's train our model!

You will be able to generate a .cc file which is your compressed model.

TinyML Detective Time!

We've *deliberately* introduced a small mistake in one line under the **"Build & Train the Model"** section. Your mission, should you choose to accept it, is to **find the incorrect line and fix it** before training your model. 🔍

Deploy the Model

1. Use a USB cable to connect the Arduino Nano 33 BLE Sense to your machine. You should see the green LED power indicator come on when the board first receives power.
2. Open the lab8-classifier.ino sketch, which you can find in the Slack channel.



Deploy the Model

3. As always, use the Tools drop-down menu to select the appropriate Port and Board, and use the rightward arrow to upload / flash the code.
4. Now, open the serial monitor and test out your custom model. As a reminder, the serial monitor will output two words as well as a confidence score for each of the gestures. The confidence score indicates how strongly the model believes you performed the gesture. Do note that every time you move the board and then stop, a new gesture will be processed, so don't be surprised to get some odd results as you move the board to prepare for a gesture.

